CSC 4880 – Artificial Intelligence

Directions: You do not need to turn in this homework. It is meant as a study guide, along with the in-class worksheets, for questions that will appear in the quiz, possibly with different phrasings, prompt formats, or different numerical examples.

Foundations

1. **Learning settings**

Define supervised learning. Then classify each task as supervised learning, unsupervised learning, or reinforcement learning, and give one sentence of justification:

(a) Grouping customers into previously unknown behavior clusters.
(b) Learning to play a game from observations and rewards collected by self-play.
(c) Predicting a house price from past houses with recorded sale prices.

2. **Features labels and examples**

A dataset contains rows of patients each with the following columns:

- Age
- blood pressure
- cholesterol
- a final column indicating whether a patient has a condition.

Identify the examples, features, and label. Is the target a classification or a prediction/regression target?

3. **Data splits**

Explain distinct roles for training, validation, and test sets. Why is it a problem to repeatedly inspect test accuracy while choosing features or hyperparameters?

4. **Eager and lazy learners**

Compare an eager learner and a lazy learner. For each, state when most computation occurs (training or inference) and whether the stored representation is typically smaller than the training data. Classify kNN and ID3 decision trees as eager or lazy.

5. **Desirable properties**

Name four desirable properties of a learned model other than raw predictive accuracy. For one of them, explain a concrete situation where it could matter more than a small accuracy advantage.

6. **Overfitting diagnosis**

A model achieves 99% accuracy on training data, 82% on validation data, and 81% on test data. What problem is suggested? Give two actions, grounded in the methods from this unit, that could plausibly improve generalization.

k Nearest Neighbors

7. **Mechanics of kNN**

Describe how kNN makes a prediction for

(a) a discrete target and
(b) a real-valued target.

What role does k play?

8. **Hand classification with kNN**

Consider the following labeled training points:

Point ID	X1 coordinate	X2 coordinate	Label
A	0	0	Red
B	2	0	Blue
C	0	3	Blue
D	3	3	Red

Use Euclidean distance and unweighted majority vote. For query Q=(1,1):

(a) Rank A, B, C, D by distance from Q (exact squared distances are sufficient).

(b) Predict the label with k=1 and with k=3.

(c) Explain why an odd k is often convenient in binary classification.

9. **Choosing k and noise**

Compare k=1 with a substantially larger k. Which is more sensitive to a mislabeled point near the query? Which is more likely to smooth away a small genuine local region? Relate your answer to our discussion of bias and variance

10. Weighted neighbors

Suppose a query has three nearest neighbors: Red at distance 0.2, Blue at 0.4, and Blue at 2.0. Under ordinary majority vote, what label is predicted? Explain qualitatively how distance-weighting could change the result, without selecting a particular weighting formula.

Decision Trees

You may use the following formulas for questions in this section. To avoid reliance on calculators, a numerical reference will be given for questions as needed.

i. Entropy of a node with set of examples S with k distinct classes:

$$H(S) = -\sum_{c=1}^{k} \log_2 p_c$$

ii. Entropy of a node with binary label where p is the proportion of the **YES** class:

$$H(S) = -p \log_2 p - (1-p) \log_2 (1-p)$$

iii. If attribute A has v possible values and partitions dataset D into subsets $D_1, D_2, \ldots, D_v$

$$\sum_{c=1}^{k} \quad \text{and} \quad \sum_{i=1}^{v}$$

$$Info_A(D) = \sum_{i=1}^{v} H(D_i)$$

$$Gain(A, D) = H(D) - Info_A(D)$$

11. Reading a tree

A tree first predicts whether you should go outside (Yes/No) based on **Weather** (Rain/Sun) and **Wind** (Strong/Weak). Suppose the tree first checks **Weather**:

- If **Weather** = Rain it tests **Wind** and predicts No if Strong, Yes if Weak.
- If **Weather** = Sun, it predicts Yes.

(a) Classify (Rain, Strong).

(b) Classify (Sun, Strong).

12. Entropy warm up

Consider the two nodes below, each with 8 examples

- Node 1: 4 positive, 4 negative.
- Node 2: 8 positive, 0 negative.

Compute the entropy of each node. Which node is more pure, and why?

You may use $\log_2\left(\frac{1}{2}\right) = -1$ and the convention $0\log_2(0) = 0$

13. Information gain by hand

A dataset D has 8 examples: 4 Yes and 4 No, so Info(D)=1 bit. Consider two binary attributes:

Attribute A: A=0 contains 3 Yes, 1 No; A=1 contains 1 Yes, 3 No.

Attribute B: B=0 contains 4 Yes, 0 No; B=1 contains 0 Yes, 4 No.

(a) Compute $Info_A(D)$ and $Gain(A, D)$
(b) Compute $Info_B(D)$ and $Gain(B, D)$
(c) Which attribute does ID3 select at this node? Interpret the result in terms of child purity.

You may use H(3/4,1/4) approximately 0.811.

14. Importance of weighted entropy

Consider a set with 100 examples. A binary attribute isolates 1 positive example in one perfectly pure branch; its other branch contains the remaining 99 examples, still with nearly the original class mix. Explain why the post-split entropy must weight branches by their relative sizes. What would go wrong if it simply averaged each branch entropy equally?

15. ID3 stopping cases

State the prediction returned by ID3 in each case:

(a) All examples at a node have label Yes.

(b) No attributes remain, but labels are mixed.

(c) A branch has no examples after splitting.

Also give one reason an implementation might stop before these cases.

16. Regression trees

How does a regression tree differ from the classification tree presented in ID3 with respect to:

(a) the output at a leaf

(b) the notion of goodness of a split

(c) tests for real-valued features?

Perceptron and Linear Classification

17. Decision boundary

A perceptron has w=(2,-1) and b=-3 and predicts +1 when $h = w^{\top}x + b > 0$, otherwise -1.

(a) Write the equation of its decision boundary.

(b) Predict the class of x=(2,0).

(c) Predict the class of x=(0,0).

18. Geometry of a perceptron

Explain the geometric meaning of $w^{\top}x + b = 0$ in two dimensions. What does the sign of $w^{\top}x + b$ tell you? What is the role of the bias b?

19. Perceptron update

Use learning rate $\eta = 1$. Suppose the current parameters are w=(0,1), b=0. A training point x=(-1, 1/2) has y=-1.

(a) Compute h and y*h. Is the point correctly classified under the lecture rule?

(b) Apply the update rule $w \leftarrow w + \eta(x * y)$ and $b \leftarrow b + \eta y$.

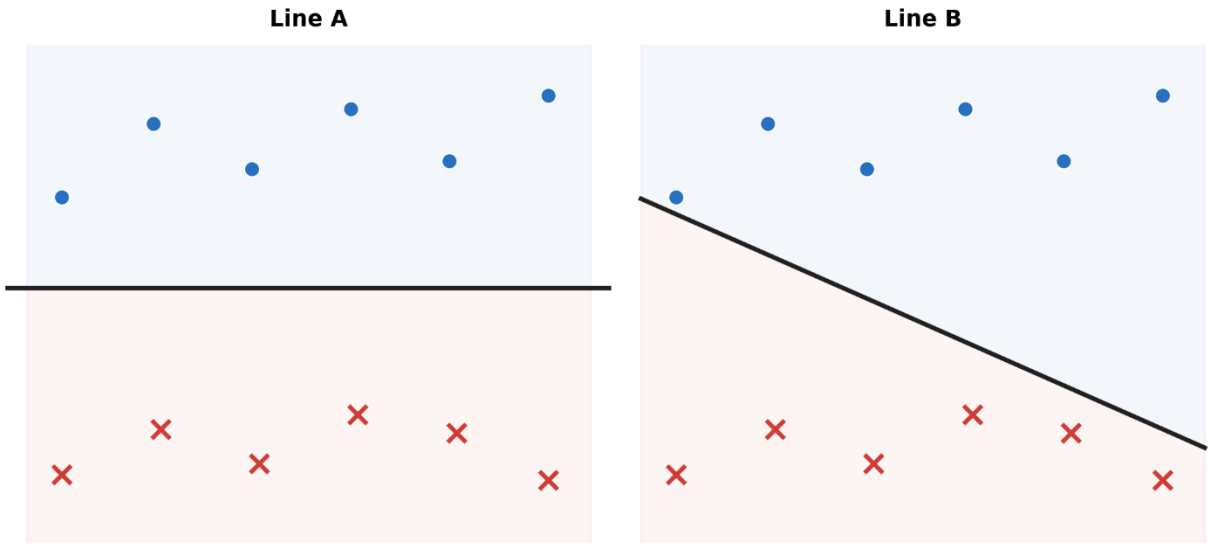
(c) Verify the new activation h for the same point and say whether the point is now classified correctly

20. Separable data and limitations

What does it mean for a binary dataset to be linearly separable? Draw a small dataset that is not linearly separable. Can a perceptron ever achieve correct classification on all points in that dataset? What architectural change discussed in lecture can address this?

21. Margin and generalization

Consider the two learned decision boundaries below on the same training set. Labels are blue dots (positive) versus red crosses (negative)



a) Do both lines classify all training points correctly?
b) Which one has a wider separation margin?
c) Which decision line would be preferable if you were concerned that the training set contained small measurement errors or that the distribution data used in deployment could differ slightly from your sampled training set, and why?
d) Suppose a perceptron reaches line B as a decision boundary after a few updates. Does the standard Perceptron algorithm have any mechanism in place to detect this issue or eventually update the decision line to one closer to line A?
e) Which method mentioned in class actively seeks a separator with wide margin?